

Managing human resources is about to become easier

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HUMAN RESOURCES (HR) is a poorly named department. It usually has few resources other than overworked staff, clunky technology and piles of employee handbooks. Hassled recruiters have to sort through reams of applications that vastly outnumber the jobs available. For example, Johnson & Johnson (J&J), a consumer-goods company, receives 1.2m applications for 25,000 positions every year. AI-enabled systems can scan applications far more quickly than humans and work out whether candidates are a good fit.

Oddly enough, they may also inject more humanity into hiring. According to Athena Karp of HiredScore, a startup that uses algorithms to screen candidates for J&J and others, only around 15-20% of applicants typically hold the right qualifications for a job, but they are rarely told why they were not hired, nor are they pointed to more suitable jobs. Technology is helping “give respect back to candidates”, she says.

Nvidia, a chipmaker, also gets more résumés than it can comfortably cope with, so it spent a year building its own system to predict which candidates are worth interviewing. It has recognised

patterns that recruiters might not: for example, candidates who submit especially long résumés turn out to do less well than others, so those extra words will count against them. Hilton, a hotel chain, has shortened the average time it takes to hire a candidate from 42 days to five with the help of HireVue, a startup. It analyses videos of candidates answering questions and uses AI to judge their verbal skills, intonation and gestures. This can be especially helpful when the candidate comes from a different culture or speaks another first language, says Ellyn Shook, chief leadership and HR officer of Accenture, a consultancy with 435,000 employees that also uses HireVue.

Employers tend to hire candidates who are like themselves, which makes for undiversified workplaces. Orchestras, for example, used to be mostly male. Recruitment of female musicians went up only when they introduced “blind” auditions behind a screen.

Algorithms can act as virtual screens, making hiring fairer.

Pymetrics, a startup whose clients include companies such as Unilever, a consumer-goods giant, and Nielsen, a research firm, offers a set of games for candidates to play, usually at an early stage of the recruitment process, that ignore factors such as gender, race and level of education. Instead they test candidates for some 80 traits such as memory and attitude to risk. Pymetrics then uses machine learning to measure applicants against top performers and predict their suitability for a role. This can help candidates without conventional qualifications.

Another firm that is helping companies become more diverse is Textio, a startup that uses AI to improve job descriptions. For example, it has found that corporate jargon like “stakeholders” and “synergies” tend to drive away certain candidates, especially non-whites, and that women are less likely to apply for a job that is described as “managing” than “developing” a team. Tweaking job descriptions can get 25% more qualified people through the door

and boost recruitment among minorities, says Kieran Snyder, Textio's boss.

Another time

Recruiters often come across candidates who have good qualifications but are not the right fit for the particular position they are trying to fill. In the past, there was no way of redirecting them to other jobs as they became available. AI will make it possible to “repurpose candidates we have attracted before”, says Sjoerd Gehring, vice-president of talent acquisition for J&J. The health-care giant uses HiredScore, a startup, to grade candidates. When a vacancy opens up, the system automatically generates a shortlist of candidates that could be a good fit. This will bring big cost savings, says Mr Gehring.

AI can also help with managing employees. HR professionals and recruiters at big firms cannot possibly know all their own talented workers across countries and departments, says Chris Louie of Nielsen. His company is using AI to improve internal mobility. Twine Labs, a startup that is working with Nielsen, suggests internal candidates for new roles, based on employee data and job requirements, taking in hundreds of variables. Around half the candidates it suggests are approved and promoted, says Joseph Quan, Twine Labs' boss. That is about the same success rate as for a human recruiter.

Another use for AI is to help employers reduce staff turnover. On average, replacing a worker takes around 20% of annual salary, sometimes much more. Workday, a software firm, has started to predict how likely employees are to leave. It looks at around 60 factors—such as pay, time between holidays taken and turnover in managers to whom the employee reports—and flags those at risk of quitting so companies can try to retain them.

Arena, a startup that works with hospitals and care-home companies, where turnover is high, considers retention even before

it takes someone on. By using data from job applications and third parties to predict which applicants are likely to stay for more than a year, Arena has reduced its clients' median turnover by 38%, says Michael Rosenbaum, Arena's boss.

In future AI may also be used to determine pay. Infosys is looking into using AI to decide when to give employees a rise, based on their performance and their pay relative to that of colleagues. The technology will make pay fairer by taking biases and personality traits out of consideration, says Sudhir Jha, head of product management and strategy at Infosys. But there is a risk that workers will try to game the system.

All this points to a broader issue in AI: transparency. Companies will need to ensure that algorithms are being constantly monitored. In America, where it is illegal to discriminate against protected groups such as racial minorities, firms must be able to prove that they are hiring from these groups roughly in proportion to the population and are not introducing any bias, says Mr Rosenbaum. Startup bosses say they offer their clients transparency and regularly check their algorithms to make sure they are free of bias. But as AI becomes more prevalent, concerns will grow that algorithms could reinforce discrimination.

Recruitment is just one example of the technological disruption that AI will bring to the workforce. The number of recruiters will come down, because AI will handle many of the mundane tasks they used to do, and face-to-face interviews will become rarer. At Unilever only shortlisted candidates are now interviewed, after several rounds of AI-enabled screening and recorded interviews through HireVue. For the remaining recruiters, though, AI will make work easier and more interesting.

It may even help some of the workers it displaces. Accenture is rolling out a custom-built tool called Job Buddy which tells employees how vulnerable their job is to automation and predicts

what training they might need so they can develop the right skills for the future. Ms Shook of Accenture says that around 80% of the people who have tried it are taking the advice it offers. But they may not have much choice.